Ipek Naz Obek

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Education

B.A., Cognitive Science — University of Pennsylvania, Philadelphia, PA — May 2025
Concentration: Computation and Cognition
Minors: Data Science, Computational Neuroscience
Summa Cum Laude; Distinction in Cognitive Science
Benjamin Franklin Scholar; Certificate in French
Honors Thesis: *Inferring Level Progress in Goal-Directed and Open-Ended Tasks*
Advisors: Dr. Alison Gopnik, Dr. Sharon Thompson-Schill, Annya Dahmani
Relevant Coursework: Programming Languages, Probability, Data Mining, Statistical Inference, Calculus I & II, Linear Algebra, Cognitive & Computational Neuroscience, Judgment & Decisions, Behavioral Economics, Developmental Psychology

Academic Positions

Research Technician and Lab Manager — Boston University, Boston, MA — Aug 2025–present
Computation, Learning, and Development Lab
PI: Dr. Kate Nussenbaum

Research Assistant — UC Berkeley, Berkeley, CA — Jun 2024–May 2025
Gopnik Cognitive Development and Learning Lab
PI: Dr. Alison Gopnik

Honors Thesis Student — University of Pennsylvania, Philadelphia, PA — Jan 2024–May 2025
Thompson-Schill Lab
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

Research Assistant — University of Pennsylvania, Philadelphia, PA — Jun–Sep 2023
Center for Neuroengineering and Therapeutics
PI: Dr. Brian Litt

Research Experience

Computation, Learning, and Development Lab — Boston University, Boston, MA — Aug 2025–present
Research Technician and Lab Manager
PI: Dr. Kate Nussenbaum

The development of goal commitment (in progress; with Eleanor Holton): examines how goal pursuit strategies, and attention to current goals, change from childhood to adulthood.

- Adapted and programmed a goal-pursuit task (Holton et al., 2024) in jsPsych for online testing across development.
- Fit and compared computational models of alternative persistence strategies in Python, and examined how attention to current goals relates to decisions to persist or switch across ages.
- Analyzed trial-level choice data with mixed-effects models in R; presented at Flux 2026, manuscript in preparation.

Sunk costs and persistence (in progress; with Dr. Brian Sweis): a translational study adapting a rodent foraging paradigm to ask how time already invested shapes decisions to persist or quit.

- Designed and programmed a human version of a rodent foraging paradigm (the Web-Surf Task) in jsPsych, in which participants decide whether to wait for videos to load or skip to the next offer under a limited time budget.
- Tested participants aged 8–25 recruited through the lab's participant database and Prolific, and analyzed choice data in R.
- Refining the task by introducing unexpected changes in waiting time to elicit more informative quitting decisions.
- Coordinate a cross-disciplinary collaboration between the Nussenbaum and Sweis labs.

Learning progress and exploration (in progress; with Dr. Julia Leonard): an independent project testing whether exploration is guided by learning progress, by progress relative to one's expectations, or both.

- Designing an experiment in which participants predict their performance across environments that differ in learnability before choosing where to explore.
- Running model simulations to verify that the design can separate the contributions of learning progress and learning-progress prediction errors to choice.

Grit as a potential moderator of susceptibility to sunk costs (in progress; with Jennifer Fu): tests whether people higher in grit are more likely to persist because of time already invested, using the Web-Surf Task.

- Mentoring an undergraduate, Jennifer Fu, through her independent research project, from study design to data collection on Prolific.
- Meeting weekly to guide her in programming the task and analyzing data on how invested time, remaining time, and grit relate to quitting decisions.

Environmental volatility and persistence (in progress; undergraduate honors thesis): asks whether prior experience with volatile environments influences decisions to persist or give up.

- Mentoring an undergraduate through an honors thesis on this question, guiding study design, programming, and analysis.

Lab management

- As the lab's first lab manager, established experimental, recruitment, and data-collection infrastructure, recruiting over 1,000 participants into the lab's database for developmental studies.
- Wrote and organized the lab's foundational documentation, including manuals and task lists for research technicians, lab managers, graduate students, and undergraduates, and scripts for paying participants and contacting partner organizations.
- Lead recruitment with undergraduate research assistants, reaching out to newsletters, schools, academic clubs, and community organizations; design recruitment posters and videos and manage the lab's social media accounts and paid advertising.
- Conduct weekly verification calls with participants enrolling in the lab's studies.
- Organize a biweekly multi-lab journal club bringing together researchers to discuss current papers.

Gopnik Cognitive Development and Learning Lab UC Berkeley, Berkeley, CA Jun 2024–May 2025

Research Assistant

PI: Dr. Alison Gopnik; Advisor: Annya Dahmani

Exploration and learning progress in early childhood (with Annya Dahmani): examines how young children choose between familiar activities and new challenges, and how they track their own learning progress.

- Collected behavioral data in person at museums in San Francisco for 3 interactive virtual studies on exploration and learning progress with children aged 3–8.
- Recruited families on-site at museums every week, increasing the lab's recruitment rate.
- Wrote analysis scripts for the lab's data pipelines, including coding participants' responses as exploratory or exploitative choices.
- Designed and analyzed experiments examining how people construct and evaluate curricula for complex, multi-step tasks.
- Prepared weekly lab presentations on background literature.

Thompson-Schill Lab — University of Pennsylvania, Philadelphia, PA — Jan 2024–May 2025
Honors Thesis Student
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

Inferring learning progress in humans and machines (honors thesis; co-advised by Dr. Alison Gopnik and Annya Dahmani): asks how people infer how much someone has learned from their choices, and whether AI models make the same inferences.

- Conducted an extensive literature review to identify available benchmarks, datasets, and evaluation methods for measuring learning progress.
- Designed a novel behavioral paradigm in a Procgen environment (the Leaper game), in which participants saw which level a player started on and chose next, then judged how much of the first level the player had completed.
- Tested online human participants and a large language model (GPT-4o) on the same task.
- Manipulated whether observers knew the learner's goal, and compared human and GPT-4o judgments of learning progress across goal-directed and open-ended conditions.
- Participated in weekly joint lab meetings, discussing current research and literature with faculty and lab members.

Center for Neuroengineering and Therapeutics — University of Pennsylvania, Philadelphia, PA — Jun–Sep 2023
Research Assistant
PI: Dr. Brian Litt; Advisor: Dr. James Gugger

Lesion characteristics and post-traumatic epilepsy (MindCORE summer fellowship): examines how the size and location of traumatic-brain-injury lesions relate to later epilepsy.

- Developed a preprocessing pipeline for clinical neuroimaging data from over 100 participants.
- Wrote Python scripts that automated labeling of lesion location and size in each brain, reducing reliance on manual annotation.
- Developed quantitative and machine-learning analyses in Python relating lesion volume and location to post-traumatic epilepsy outcomes.

Honors & Awards

Distinction in Cognitive Science — University of Pennsylvania — 2025
Awarded for completing an honors thesis, "Inferring Level Progress in Goal-Directed and Open-Ended Tasks," and for academic excellence in the major. Thesis advised by Alison Gopnik and Annya Dahmani (UC Berkeley Psychology) and Sharon Thompson-Schill (Penn Psychology).

Summa Cum Laude — University of Pennsylvania — 2025
Graduated with highest Latin honors from the College of Arts & Sciences.

Benjamin Franklin Talk Prize — Benjamin Franklin Scholars, University of Pennsylvania — 2025
Recognized for the best senior research presentation among the 2025 cohort of Benjamin Franklin Scholars, Penn's interdisciplinary honors program.

Lucid Communication Challenge, Engineering and Physical Sciences Talk — Penn National Research Conference — 2024
Selected from a field of applicants at the National Research Conference at Penn to present my honors thesis research as a talk.

Dean's List — University of Pennsylvania — 2021–2025
Awarded in every academic year for a GPA of 3.7 or higher.

Benjamin Franklin Scholar — University of Pennsylvania — 2021–2025
Member of Penn's interdisciplinary honors program in the liberal arts and sciences.

Grants & Fellowships

Pinkel Research Fund — University of Pennsylvania — 2025
Research funding awarded to support my senior honors thesis on curriculum learning in humans and machines.

Benjamin Franklin Research Grant — Benjamin Franklin Scholars, University of Pennsylvania — 2024

Summer research funding awarded by the Benjamin Franklin Scholars Program; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

Career Services Research Grant — Penn Career Services — 2024

Competitive summer funding awarded by Penn Career Services; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

MindCORE Fellowship — MindCORE, University of Pennsylvania — 2023

Selected for a fully funded 10-week summer research program in cognitive science and neuroscience at Penn's hub for the integrative study of the mind.

Undergraduate Research Mentoring Award — Penn Undergraduate Research Mentoring Program — 2023

Fellowship awarded through Penn's Undergraduate Research Mentoring Program, which funds first- and second-year students to conduct summer research with a faculty mentor.

Manuscripts in Preparation

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (in preparation). *Examining the development of goal commitment*.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (in preparation). *Level learning progress*.
- Fu, J., **Obek, I. N.**, & Nussenbaum, K. (in preparation). *Grit as a potential moderator of susceptibility to sunk costs in decision-making tasks*.

Unpublished Manuscripts

- **Obek, I. N.** (2025). *Inferring level progress in goal-directed and open-ended tasks*. Honors Thesis, University of Pennsylvania. Advised by A. Gopnik, S. Thompson-Schill, and A. Dahmani.
- **Obek, I. N.**, & Flombaum, J. (2021). *Are empathy and concern psychologically distinct, and is this distinction intuitive?* Research paper, Pioneer Academics Research Program.

Presentations

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (2026, September). *Examining the development of goal commitment* [Poster]. Flux Society for Developmental Cognitive Neuroscience, La Jolla, CA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Poster]. Curiosity, Information Seeking, and Exploration Conference, Brown University, Providence, RI.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Talk]. Big Team Science Conference.
- **Obek, I. N.** (2025, April). *Machines that think like kids* [Talk]. Benjamin Franklin Scholars, University of Pennsylvania, Philadelphia, PA. Awarded the Benjamin Franklin Talk Prize.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Talk]. Lucid Communication Challenge, Engineering and Physical Sciences session, Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, Fall). *Automated curriculum learning in children and machines* [Poster]. Center for Undergraduate Research and Fellowships (CURF) Poster Symposium, University of Pennsylvania, Philadelphia, PA.

- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Poster]. Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, Fall). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. Center for Undergraduate Research and Fellowships (CURF) Poster Symposium, University of Pennsylvania, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, August). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. MindCORE Research Fellowship Conference, Philadelphia, PA.

Teaching

Neuroscience and Society — Teaching Assistant, Penn Undergraduate Program in Neuroscience — 2024–2025
Instructor: Dr. Sharon Thompson-Schill
Supported students through weekly office hours, in-class instruction, and group activities, and provided individualized academic guidance.

Introduction to Cognitive Science — Teaching Assistant, Penn Undergraduate Program in Cognitive Science — Fall 2023
Instructor: Dr. Russell Richie
Led weekly recitations and office hours for 20+ students, developed instructional materials, and guided students through course concepts and assignments.

MATH 114: Calculus II — Tutor, Penn Weingarten Tutoring Center — Fall 2022
Tutored first-year students for 4 hours per week, providing individualized support with mathematical problem-solving and study strategies.

Mentorship

Benjamin Franklin Scholars Mentor Program — Peer Mentor, University of Pennsylvania — 2024–2025
Mentored 7 Benjamin Franklin Scholars in their academic and social transition to Penn through regular individual outreach and community-building events.

Center for Undergraduate Research and Fellowships — Research Peer Advisor, University of Pennsylvania — 2024–2025
Advised undergraduates pursuing research opportunities and organized research programming, including an international faculty symposium on the experiences and challenges of international academics.

Pioneer Academics — Program Coordinator and Advisory Board Member — 2022–2025
Advised 100+ students through the research and writing process and coordinated research seminars on effective research practices and faculty mentorship.

Young Guru Academy — Alumni Mentor — 2019–present
Mentor Turkish high school and university students pursuing international education and STEM careers. Led STEM outreach across 5+ cities in rural Anatolia, training teachers and students and delivering hands-on science education to underserved communities; presented at YGA's annual summit to ~20,000 attendees.

Leadership & Service

Cognitive Science Student Association — Founder & President, University of Pennsylvania — 2024–2025
Founded and led Penn's Cognitive Science Student Association, organizing faculty talks, research seminars, mentoring, and networking initiatives connecting students with faculty, alumni, and industry professionals.

The DREAM Program — Recruitment Chair, Philadelphia, PA — 2023–2025

Led volunteer recruitment and weekly educational programming for preschool-aged children in West Philadelphia, including hands-on science activities designed to foster curiosity, collaboration, and confidence in learning.

Turkish Student Association — President, University of Pennsylvania — 2022–2025

Directed a 50-member organization and community initiatives, including raising $8K for earthquake-relief efforts.

Assembly of International Students — Social Committee Member, University of Pennsylvania — 2021–2025

Organized social and cultural programming to foster community among Penn's international student population.

The Daily Pennsylvanian — Photographer, University of Pennsylvania — 2021–2025

Photographed campus events and contributed visual content for multimedia stories at Penn's student newspaper.

Skills

Programming: Python, R, JavaScript, MATLAB, SQL, Java, OCaml

Research tools & platforms: jsPsych, PsychoPy, Qualtrics, Prolific, Pavlovia, Git/GitHub, Jupyter, LaTeX

Neuroimaging: fMRI preprocessing and GLM analysis scripts in AFNI (afni_proc.py), region-of-interest analysis, MRI lesion segmentation pipelines in Python; trained in fMRI study design and scanner operation (PSYC 4230: Introduction to fMRI)

Certifications: CITI Training (Biomedical Responsible Conduct of Research; Human Subjects Research; Social-Behavioral Research with Children)

Languages: Turkish (native), English (fluent), French (fluent)

References

Dr. Kate Nussenbaum
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Dr. Alison Gopnik
Professor of Psychology
University of California, Berkeley
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Dr. Sharon Thompson-Schill
Christopher H. Browne Distinguished Professor of Psychology; Co-Director, Center for Cognitive Neuroscience
University of Pennsylvania
sschill@psych.upenn.edu

Dr. Russell Richie
Director of Strategy and Impact
Progress and Poverty Institute
Formerly Associate Director, Cognitive Science and MindCORE, University of Pennsylvania
drussellmrichie@gmail.com

Annya Dahmani
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